Probability and Statistics Assignment

Probability, inference, testing, count data and likelihood (Weeks 1–3)

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Due date: as announced in class Total: 70 marks (+5 bonus)

Instructions. Answer every question in Parts A–G; Question 19 is an optional bonus. Show your working: most of the marks are for the reasoning and for saying what a number means, not for the number alone. For Parts A–F you may use a calculator, or `pnorm`, `pt`, `pchisq` (R) or `scipy.stats` (Python), to look up probabilities and quantiles; a table of common critical values is on the last page. Every other step must be done by hand. Part G is a computing task: submit a script or notebook in Python or R that runs from start to finish on the data file `cytokine_il6.csv`, and state the random seed you used. Discussing ideas with classmates is encouraged, but the work you submit must be your own, so name anyone you worked with. If you used an AI assistant, add a short note saying what you used it for. Give numerical answers to four significant figures unless stated otherwise. All data are simulated or illustrative.

Part	Topic	Marks
A	Probability basics (Questions 1–4)	12
B	Sampling, standard errors and confidence intervals (Questions 5–7)	11
C	Hypothesis testing (Questions 8–11)	13
D	Multiple testing and Bayes' theorem (Questions 12–13)	8
E	Count data (Questions 14–15)	8
F	Likelihood (Questions 16–17)	10
G	Computing: a two-group comparison done three ways (Question 18)	8
Bonus	Question 19	5

Part A Probability basics

Question 1

[3 marks]

In a proteomics experiment, 500 proteins were quantified. Of these, 150 are membrane proteins, 100 were found to be phosphorylated, and 45 are both membrane proteins and phosphorylated. One protein is picked at random.

(a) What is the probability that it is a membrane protein *or* phosphorylated (or both)?

(b) Given that it is a membrane protein, what is the probability that it is phosphorylated?

(c) Are "membrane protein" and "phosphorylated" independent events? Justify your answer with numbers, and say in words what you find.

Question 2 [3 marks]

A CRISPR knockout succeeds in each isolated clone independently with probability 0.7. You isolate 6 clones.

✓ (a) What is the probability that exactly 4 clones carry the knockout?

✓ (b) What is the probability that at least one clone carries it?

✓ (c) Find the expected number of successful clones and its variance.

✓ (d) What is the probability that at least 5 clones carry the knockout?

Question 3 [3 marks]

The number of plaques on a phage plate follows a Poisson distribution with mean 3.2.

✓ (a) What is the probability that a plate has no plaques?

✓ (b) What is the probability that a plate has at most 2 plaques?

✓ (c) Five plates are counted independently. What is the distribution of the total number of plaques, and what are its mean and standard deviation?

(d) A student's five plates have mean 3.2 but variance 9. What does this suggest about the Poisson model for these plates?

$3.2 \pm \underline{5}$

Question 4 [3 marks]

A protein's concentration in plasma, X, is normally distributed across healthy individuals with mean 50 μg/mL and standard deviation 8 μg/mL.

(a) What proportion of healthy individuals have a concentration above 62 μg/mL?

(b) Find the concentration that only 2.5% of healthy individuals exceed.

(c) Four healthy individuals are chosen at random, and $\bar{X}$ is the mean of their four concentrations. What are the mean and standard deviation of $\bar{X}$, and what is the probability that $\bar{X}$ exceeds 62 μg/mL? Compare your answer with (a).

$\sigma = 8\,\mu g/ml$

$\mu = 50\,\mu g/ml$

$\mu + 2\sigma =$

Part B Sampling, standard errors and confidence intervals

Question 5

[5 marks]

The primary root length (mm) of ten *Arabidopsis* seedlings grown on a new medium was

$$21.4,\quad 18.9,\quad 23.1,\quad 20.5,\quad 19.8,\quad 22.6,\quad 24.0,\quad 20.1,\quad 21.7,\quad 19.3.$$

(a) Compute the sample mean, the sample standard deviation and the standard error of the mean. Explain in one sentence the difference between the standard deviation and the standard error.

(b) Construct a 95% confidence interval for the mean root length, stating the multiplier you used and why.

(c) On the standard medium the mean root length is known to be 20.0 mm. Without doing a separate test, say what your interval implies about whether the new medium changes root length at the 5% level.

(d) Now carry out the one-sample t-test of $H_0 : \mu = 20.0$ and confirm that it agrees with (c).

(e) If you grew 40 seedlings and the standard deviation stayed the same, roughly how wide would the 95% interval be? Why is it not exactly half as wide as with 10 seedlings?

Question 6

[3 marks]

In a single-cell RNA-seq experiment, the number of reads per cell is strongly right-skewed, with mean 5,000 and standard deviation 4,000.

(a) For a random sample of 64 cells, use the central limit theorem to find the approximate probability that the sample mean exceeds 6,000 reads.

(b) Explain why the normal approximation is reasonable for the mean of 64 cells but would be a poor way to find the probability that a *single* cell has more than 6,000 reads.

(c) How many cells would you need for the standard error of the mean to be at most 200 reads?

Question 7

[3 marks]

In a pooled screen, 36 of 120 sgRNAs produced a visible phenotype.

(a) Estimate the proportion of sgRNAs that produce the phenotype, its standard error, and a 95% confidence interval using the normal approximation $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$.

(b) How many sgRNAs would be needed for this interval to have a margin of error of ±0.05, assuming the true proportion is near 0.3?

(c) Suppose instead that only 2 of 120 sgRNAs had produced the phenotype. Compute the same interval and explain what goes wrong.

Part C Hypothesis testing

Question 8

[4 marks]

The expression of a gene ($\log_2$ normalised units) was measured in tumour tissue and in adjacent normal tissue from the same six patients:

Patient	1	2	3	4	5	6
Normal	7.2	6.8	8.1	7.5	6.9	7.7
Tumour	8.0	7.1	8.9	7.9	7.8	8.0

(a) State the null and alternative hypotheses for a paired test of whether expression differs between tumour and normal tissue.

(b) Carry out the paired t-test: give the mean difference, its standard error, the t statistic, the degrees of freedom and a conclusion at the 5% level. Also give a 95% confidence interval for the mean difference.

(c) A colleague analyses the same numbers as two independent groups with a Welch t-test and obtains $p = 0.090$. Explain why the two analyses disagree so much, and which one is correct for this design.

Question 9

[3 marks]

Adult body mass (g) was recorded for mice on two diets:

Diet	n	Mean	SD
High-fat	8	28.4	2.1
Control	10	25.9	3.0

(a) Compute the Welch t statistic for the difference in means.

(b) Its degrees of freedom are given by the Welch–Satterthwaite formula

$$\nu = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1 - 1} + \dfrac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Compute ν, and test at the 5% level whether the diets differ.

(c) A student writes: "The diets do not differ in their effect on body mass." Rewrite this conclusion so that it is correct.

Question 10 [3 marks]

Among 80 genes with a CpG-island promoter, 52 are broadly expressed across tissues; among 70 genes without one, 28 are broadly expressed.

(a) Set out the 2×2 table and the expected counts under independence.

(b) Compute the chi-square statistic (without continuity correction), state its degrees of freedom, and give your conclusion at the 5% level.

(c) When would you use Fisher's exact test instead, and what does the odds ratio for this table tell you?

Question 11 [3 marks]

Changes in a gene's $\log_2$ expression between two conditions are measured as n independent paired differences, and these differences have standard deviation $\sigma = 1.2$. You plan a two-sided z-test at $\alpha = 0.05$ of whether the mean change is zero.

(a) If the true mean change is $\delta = 0.6$ and $n = 16$, show that the power is approximately $\Phi\big(\delta\sqrt{n}/\sigma - 1.96\big)$, and evaluate it.

(b) How many replicates are needed for 80% power? (Use $z_{0.80} = 0.8416$.)

(c) If 20 genes each truly change by 0.6 and each is tested with $n = 16$, how many of them do you expect to miss?

Part D Multiple testing and Bayes' theorem

Question 12 [4 marks]

Twelve candidate genes were tested, giving these p-values (already sorted):

$$0.0004,\ 0.0021,\ 0.0039,\ 0.0052,\ 0.011,\ 0.018,\ 0.029,\ 0.041,\ 0.087,\ 0.21,\ 0.46,\ 0.73.$$

(a) If all twelve null hypotheses were true and the tests independent, what is the probability of at least one p-value below 0.05?

(b) Which genes are declared significant by Bonferroni at family-wise error rate 0.05? By Holm's step-down procedure? Show the thresholds you compared with.

(c) Which genes does Benjamini–Hochberg declare significant at false discovery rate 0.05? Show the thresholds.

(d) In one or two sentences, explain what the BH result promises about the genes it declares significant, and how this differs from the promise Bonferroni makes.

Question 13 [4 marks]

A culture is known to be one of two cell lines. In line A, a surface marker is expressed by 80% of cells; in line B, by 30%. Before looking, both lines are equally likely. You stain 5 randomly chosen cells and 4 of them express the marker.

(a) Compute the likelihood of the observation under each line and the likelihood ratio in favour of line A.

(b) Use Bayes' theorem to find the posterior probability that the culture is line A.

(c) Explain why the binomial coefficient $\binom{5}{4}$ can be left out of both likelihoods.

(d) In your lab, line A is used in only 1 culture out of every 10. Using the odds form of Bayes' theorem, recompute the posterior probability of line A.

Part E Count data

Question 14 [4 marks]

A gene's read counts in five biological replicates sequenced to the same depth were

$$112, \quad 87, \quad 140, \quad 95, \quad 166.$$

(a) Compute the mean, the sample variance and the dispersion index $D = s^2/\bar{y}$.

(b) If the counts were Poisson, the statistic $X = (n-1)s^2/\bar{y}$ would approximately follow a χ^2_{n-1} distribution. Use it to test the Poisson model at the 5% level.

(c) Under the negative binomial model $\mathrm{Var}(Y) = \mu + \varphi\mu^2$, estimate φ by matching moments.

(d) In one or two sentences, say what would go wrong if you tested this gene for differential expression with a Poisson model.

Question 15 [4 marks]

A sample of 400 plants from a wild population was genotyped at a biallelic locus: 196 AA, 148 Aa and 56 aa.

(a) Estimate the frequency p of allele A.

(b) Compute the genotype counts expected under Hardy–Weinberg equilibrium.

(c) Carry out the chi-square test of Hardy–Weinberg equilibrium and give your conclusion at the 5% level.

(d) Estimate the inbreeding coefficient

$$F = 1 - \frac{\text{observed heterozygotes}}{\text{expected heterozygotes}},$$

and suggest one biological explanation that fits a plant population.

Part F Likelihood

Question 16

[5 marks]

In a lineage-tracing experiment, a progenitor cell divides repeatedly. At each division it stops dividing and differentiates with probability p, independently of earlier divisions. The number of divisions X up to and including the one at which it differentiates then has the geometric distribution

$$P(X = x) = (1-p)^{x-1}\, p, \qquad x = 1, 2, 3, \ldots$$

Eight progenitors gave $X = 3,\ 1,\ 4,\ 2,\ 5,\ 1,\ 2,\ 2$.

(a) Write down the log-likelihood $\ell(p)$ for n independent observations.

(b) Show that the maximum-likelihood estimate is $\hat{p} = n/\sum x_i$, and evaluate it.

(c) Find $\ell''(p)$, confirm that $\hat{p}$ is a maximum, and use $\mathrm{SE}(\hat{p}) \approx 1/\sqrt{-\ell''(\hat{p})}$ to estimate the standard error.

(d) Give an approximate 95% (Wald) confidence interval for p.

(e) Interpret $1/\hat{p}$ in biological terms.

Question 17

[5 marks]

A gene receives $y_1 = 58$ reads in a library from treated cells and $y_2 = 34$ reads in a library from control cells. The two libraries have identical sequencing depth, and each condition is represented by a single biological sample. Model the counts as independent Poisson variables with means λ_1 and λ_2.

(a) Write the log-likelihood $\ell(\lambda_1, \lambda_2)$. Find the maximum-likelihood estimates with no restriction, and under $H_0\colon \lambda_1 = \lambda_2 = \lambda$.

(b) Show that the likelihood ratio statistic is $G = 2\left[\ell(\hat{\lambda}_1, \hat{\lambda}_2) - \ell(\hat{\lambda}, \hat{\lambda})\right] = 2\sum_{j=1}^{2} y_j \ln\left(\frac{y_j}{\bar{y}}\right)$, where $\bar{y} = (y_1 + y_2)/2$.

(c) Evaluate G, compare it with the appropriate chi-square distribution, and state your conclusion at the 5% level. Explain why the chi-square distribution has one degree of freedom.

(d) Suppose this gene has a dispersion similar to the gene in Question 14. Explain why the test in (c) does not establish that the treatment changes the gene's expression.

Part G Computing: a two-group comparison done three ways

Question 18

[8 marks]

The file `cytokine_il6.csv` gives the plasma IL-6 concentration (pg/mL) in 8 control mice and 8 mice treated with an inflammatory stimulus. Write a script in Python or R that does the following, and submit it with your answers.

(a) Print the mean, median and standard deviation of each group, and plot the individual values of both groups side by side, once on the original scale and once on the log scale. Describe what you see in one or two sentences.

(b) Run a Welch t-test comparing treated with control (treated minus control) on the original scale, and again on the natural-log scale. Report t, the degrees of freedom and the p-value for each.

(c) Carry out a two-sided permutation test of the difference in mean log concentration (treated minus control), using at least 10,000 random relabellings of the 16 mice. Report the observed difference, the ratio it corresponds to on the original scale, and the permutation p-value.

(d) Use the bootstrap (at least 10,000 resamples, resampling within each group) to construct a 95% percentile confidence interval for the ratio of the median concentrations (treated / control).

(e) In three or four sentences, explain why the raw-scale and log-scale t-tests disagree, and say which result you would report and why.

Bonus

Question 19

[5 marks]

In a single-cell experiment, each of 1,000 cells receives a DNA barcode chosen uniformly at random from 100,000 possible barcodes. Two cells that receive the same barcode cannot be told apart.

(a) How many pairs of cells are there, and what is the expected number of pairs that share a barcode?

(b) Treating the number of colliding pairs as approximately Poisson, estimate the probability that no two cells share a barcode.

(c) How many possible barcodes would be needed for this probability to be at least 0.9?

Table of critical values

Standard normal	$z_{0.95} = 1.645$	$z_{0.975} = 1.960$	$z_{0.995} = 2.576$

df	5	6	7	8	9	10	12	15	16	20	39
$t_{0.975,\,\text{df}}$	2.571	2.447	2.365	2.306	2.262	2.228	2.179	2.131	2.120	2.086	2.023

df	1	2	3	4	5
$\chi^2_{0.95,\,\text{df}}$	3.841	5.991	7.815	9.488	11.070

z	0	0.25	0.5	0.75	1.0	1.25	1.5	1.75	2.0	2.5	3.0
$\Phi(z)$	0.5000	0.5987	0.6915	0.7734	0.8413	0.8944	0.9332	0.9599	0.9772	0.9938	0.9987

$\Phi(z) = P(Z \leq z)$ for a standard normal Z, and $\Phi(-z) = 1 - \Phi(z)$. For other values of z, use a calculator or software, or interpolate linearly in the table.

For a non-integer number of degrees of freedom, use the nearest smaller tabulated value (a conservative choice), or compute the exact quantile in software.

End of assignment

Theory

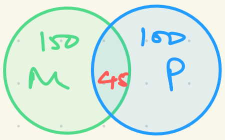
$$P(A) = \frac{\text{favourable outcome}}{\text{Total possible outcome}}$$

each outcome → equal likely

②

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

500protein → 150 membrane
→ 100 phosphorylat
→ 45 Both



ⓐ Either membrane or phosphylated → (means only M or only P) or Both

$$P(\text{membrane/phosphorylated}) = P(M) + P(P) - P(M \cap P)$$

$$P(M) = \frac{150}{500} \qquad P(P) = \frac{100}{500} \qquad P(M \cap P) = \frac{45}{500}$$

$$: P(M) + P(P) - P(M \cap P)$$

$$= \frac{150}{500} + \frac{100}{500} - \frac{45}{500}$$

$$= \frac{250 - 45}{500} = \frac{205}{500} = 0.41$$

Ⓑ

$$P(P/M) = \frac{P(P \cap M)}{P(M)}$$

Conditional Prob

$$= \frac{45/500}{150/500}$$

$$= \frac{\cancel{45}^{3}}{\cancel{150}_{10}} = 0.3$$

ⓒ Independant event mean when one event is not affected by other

$$P(P \cap M) = P(P) \times P(M)$$

$$\left(\frac{150}{500} \times \frac{\cancel{80}}{\underset{10}{\cancel{500}}}\right) = \frac{\overset{3}{\cancel{150}}}{\underset{100}{\cancel{5000}}} = 0.3$$

$$P(P \cap M) = \frac{\overset{9}{\cancel{45}}}{\underset{100}{\cancel{500}}} = \underline{0.09}$$

hence the events are dependant on each other

② clone knockout $\rightarrow$ success, $\rightarrow$ fail

only 2 possibility

success $= 0.7$ fail $= 1 - 0.7$
$= 0.3$

ⓐ that means we find out of 6 trials only 4 knockout → success

$X \rightarrow$ success

$$P(x=4) = {}^{6}C_{4}\,(0.7)^{4}\,(0.3)^{2}$$

$$= 0.324135$$

ⓑ at least 1 clone carries knockout

$$P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)+P(x=5)+P(x=6)=1$$

at least 1 knockout has happen

$$P(x=\text{atleast 1 knockout})=1-P(x=0)$$

$$=1-{}^{6}C_{0}(0.7)^{0}(0.3)^{6}$$

$$=1-(0.3)^{6}$$

$$=0.999271$$

ⓒ expectation & variance

as we are following Binomial

$$E(x)=\overset{\text{trial}}{n}\underset{\text{success prob.}}{p}$$

$$=6\times0.7$$

$$\text{variance}=np(1-p)$$

$$=6(0.7)(0.3)$$

$$=1.260$$

ⓓ

$$P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)+P(x=5)+P(x=6)=1$$

for atleast 5 clone knockout

$$P(x=5)+P(x=6)=P(x=\text{atleast 5 knock out})$$

$${}^{6}C_{5}(0.7)^{5}(0.3)^{1}+{}^{6}C_{6}(0.7)^{6}=P(x=\text{atleast 5 knock out})$$

$$0.420=P(x=\text{atleast 5 knockout})$$

③ $P(X=k) = \frac{e^{-\mu} \mu^{k}}{k!}$

ⓐ

given $\mu = 3.2$

$P(X=0) = \frac{e^{-3.2} \times (3.2)^{0}}{0!} = e^{-3.2} = 0.04076$

ⓑ

atmost 2 means plate can have $\rightarrow 0$, $\rightarrow 1$, $\rightarrow 2$

$P(X=0) + P(X=1) + P(X=2)$

$e^{-3.2} + \frac{e^{-3.2}(3.2)^{1}}{1!} + \frac{e^{-3.2}(3.2)^{2}}{2!}$

$= e^{-3.2}\left(1 + 3.2 + \frac{(3.2)^{2}}{2!}\right)$

$= e^{-3.2} \times 9.32$

$= 0.3799$

ⓒ mean & variance $= \lambda$

$E(X) = var(X) = \lambda$

since 5 plate are independant → mean can add

mean = variance $= \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5$

$= 3.2 \times 5$

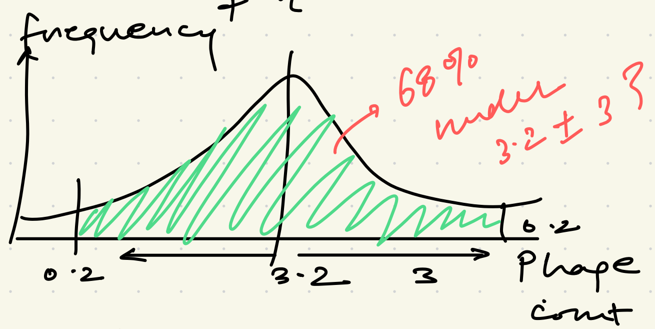
$= 16$

$\sqrt{\text{Variance}}$ = Standard devi

$\sqrt{16} = \sigma$

$+ \mu = \sigma$

ⓓ

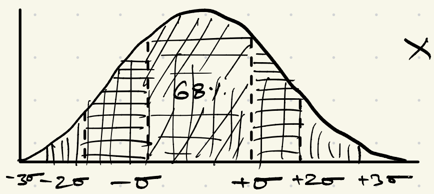


variance = 9

SD = 3

σ 2σ 3σ

68 : 95 : 99.7 .... Rule → only for normal distribution



$X \sim N(\mu, \sigma^{2})$

overdispersion?

Important

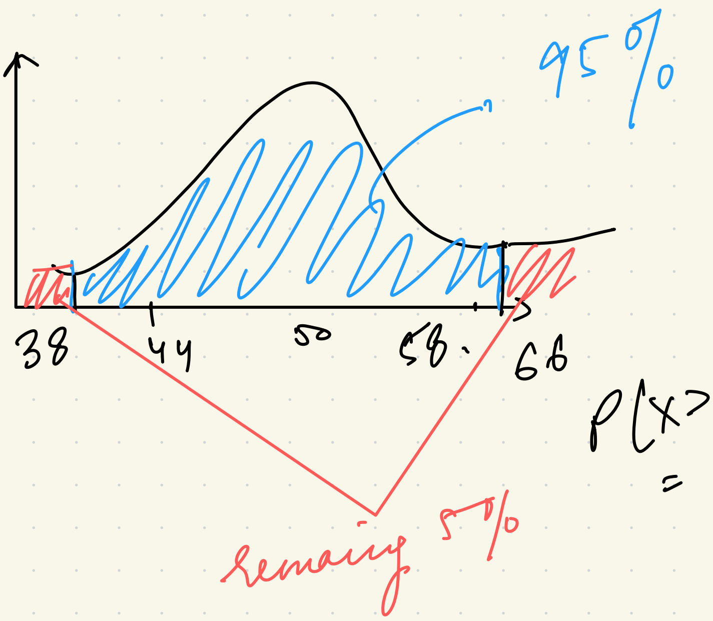
Q4 $X \sim N(\mu, \sigma^2)$

$\mu = 50 \mu g/ml$

$\sigma = 8 \mu g/ml$

Standardization

(a)



$$z = \frac{62 - 50}{8}$$

$$= 12/8$$

$$= 1.5$$

$P(X>62)$

$= 1 - Z(1.5)$

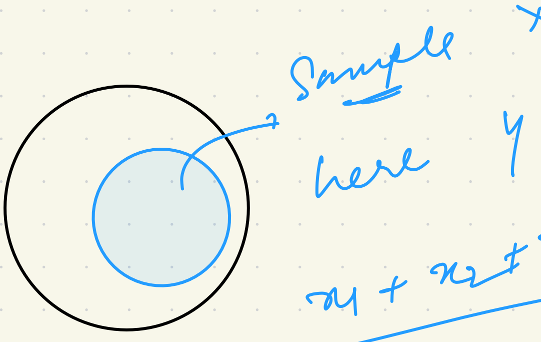
find this later

(b) concentration only 2.5% exceed is 62μg/ml ($\mu + 2\sigma$)

(c) Sample $\bar{X} =$

$\bar{x} =$

Say



$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4}{4} = \text{sample mean}$$

derive??

$\bar{x} = \mu$

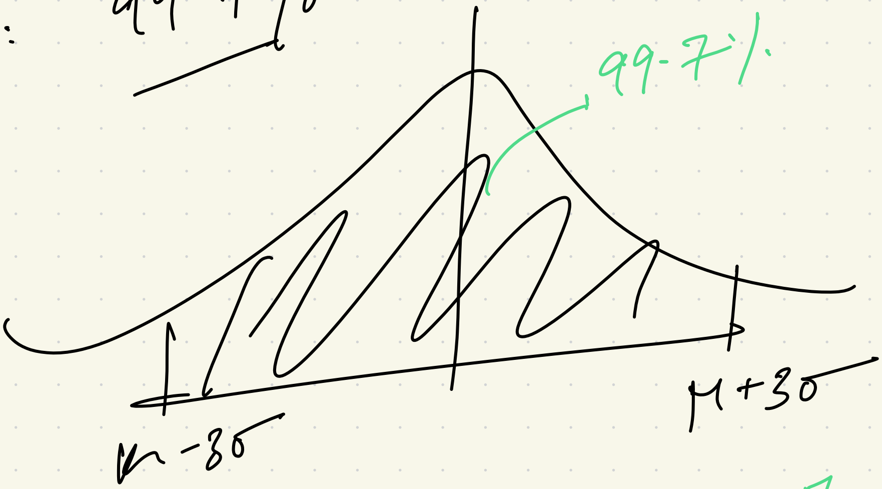
$\bar{x} = 50$

$SD(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

$$SD(\bar{x}) = \frac{8}{\sqrt{4}} = \frac{8}{2} = 4$$

$$\left(\frac{62-50}{4}\right) = \frac{12}{4} = 3$$

$(\mu \pm 3\sigma)$: $\underline{99.7\%}$



$$\frac{0.3\%}{2} = \underline{0.15\%}$$

Q5

$$\bar{x} = \frac{21.4 + 18.9 + 23.1 + 20.5 + 19.8 + 22.6 + 24.0 + 20.1 + 21.7 + 19.3}{10}$$

$$\boxed{\bar{x} = \frac{211.4}{10} = 21.14\text{mm}}$$

21.4	0.26
18.9	2.24
23.1	1.96
20.5	0.64
19.8	1.34
22.6	1.46
24.0	2.86
20.1	1.04
21.7	0.56
19.3	1.84

total

26.224

$\sigma^2 = \frac{26.224}{9} = 2.91$ ← why $n-1$?

$\sigma = 1.70$mm

error $= \frac{\sigma}{\sqrt{n}} = \frac{1.70}{\sqrt{10}} = 0.54$

$df = n-1 = 10-1 = 9$

confidence interval

had to see online

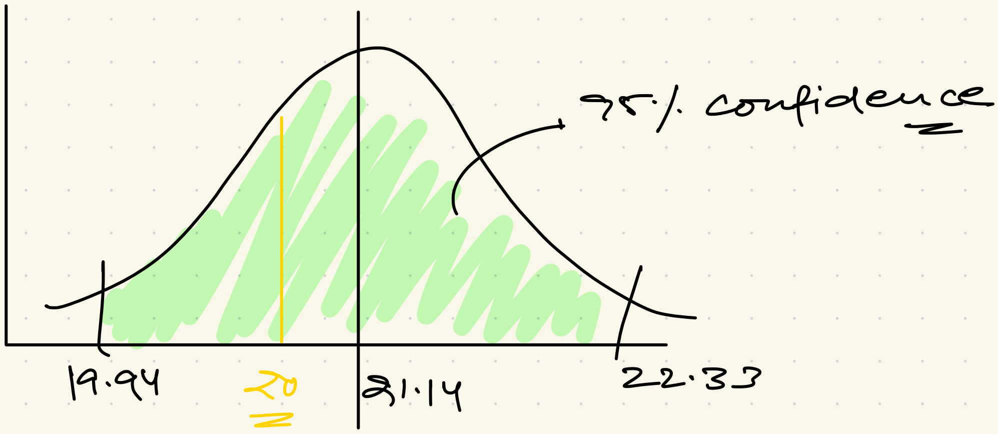
(2 side test)

$CI = \bar{x} \pm \underline{t_{0.975,9}}\left(\frac{s}{\sqrt{n}}\right)$

$CI = 21.14 \pm (2.262)(0.528)$

$(19.94 \;,\; 22.33)$ mm

ⓒ



20 lies in interval [19.94, 22.33

so it does not reject $\mu = 20$ @ 5% level

$t = \frac{\bar{X} - \mu}{\text{error}}$

ⓓ $H_0 \quad \mu = 20 \qquad H_1 \quad \mu \neq 20$

$t = \frac{21.14 - 20}{0.54} = \frac{1.14}{0.54} = 2.11$

$t_{n-1} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$df = 10-1 = 9$ for 2 side

$p = P(|T_9| \geq 2.112) = 0.06387$ $\qquad t_{0.975,⑨} = 2.262$ $\quad$ 10−1

$p > 0.05$

hence we do not reject null hypothesis

ⓔ $\sigma = 1.70$ $n = 40$

$$SE_{10} = \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{10}} \quad \text{(old)}$$

$$SE_{40} = \frac{\sigma}{\sqrt{40}} = \frac{\sigma}{2\sqrt{10}} \approx \frac{SE_{10}}{2} = \frac{0.54}{2} = 0.27$$

So new multiplyer is

$df = 40 - 1 = 39$

$$\delta_{0.975,39} \approx 2.023$$

$$\boxed{w = 2t * SE}$$

width

$2(2.023) \times (0.27) = 1.092$ mm

initial with

$2(2.262)(0.54) = 2.443$ mm

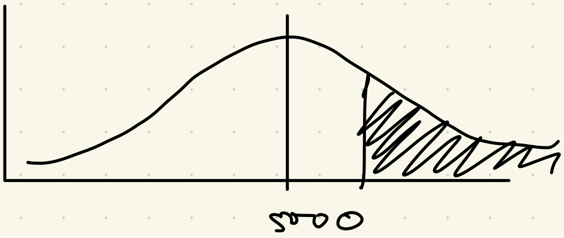
hence width decreased

Q6 ⓐ $n = 64$ $SD = 4000$

$\sigma = 4000$

$\bar{x} = 5000$

$$\text{error} = \frac{\sigma}{\sqrt{n}} \rightarrow \frac{4000}{8}$$



$$P(\bar{x} > 6000) = 1 - \phi\left(\frac{6000 - 5000}{500}\right)$$

$$= 1 - \phi(2)$$

$$\doteq 0.022\underline{75}$$

ⓑ idk ☹ → Have to check

ⓒ $$SE = \frac{4000}{\sqrt{n}} \leq 200$$

$$\frac{4000}{200} = 20$$

$$20 \leq \sqrt{n}$$

$$(20)^2 \leq n$$

$$400 \leq n$$

$n \geq 400$ cell

Topic study.

⑦ $$\hat{p} = \frac{36}{120} = 0.3$$

$SE =$

⑧

	N	T	diff
1	7.2	8	0.8
2	6.8	7.1	0.3
3	8.1	8.9	0.8
4	7.5	7.9	0.4
5	6.9	7.8	0.9
6	7.7	8.0	0.3

ⓐ question as → whether expression differs ??
So if population mean of difference = T − N
Say $T - N = D$

So null hypothesis $\underline{H_0 : \mu_d = 0}$

$H_1 : \mu_d \neq 0$

ⓑ $\Sigma d = 0.8 + 0.3 + 0.8 + 0.4 + 0.9 + 0.3$

$\Sigma d = 3.5$

mean $= 3.5/6 = 0.5833$

$\sigma^2 = (0.8 - 0.58)^2 \times 2 + 2(0.3 - 0.58)^2 + (0.4 - 0.58)^2 + (0.9 - 0.58)^2$

$= 2(0.22)^2 + 2(0.28)^2 + (0.18)^2 + (0.32)^2$

$\sigma^2 = \frac{0.388}{6} = 0.0647$

$\sigma = 0.2544$

$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.2544}{\sqrt{6}} = \frac{0.2544}{2.45} = 0.1031$

$t = \frac{\text{observed mean} - \text{null mean}}{SE}$

observed $= \frac{0.5833 - 0}{0.1031} = 5.6$

$df = n - 1 = 5$

$t_{97.5,5} = 2.571$

$5.6 > 2.571$

@ $t_{97.5,5}$: $p = P(T \leq -5.6) + P(T \geq 5.6)$

$p = 2(T > 5.6)$

$\simeq 0.0037$

$p < 0.05$ → reject null hypothesis

confidence interval

$$= 0.5833 \pm t_{0.975,5}(SE)$$
$$= 0.5833 \pm 2.57(0.1031)$$
$$= 0.5833 \pm 0.26$$
$$= [0.32, 0.84]$$

ⓒ Tumour and normal measurement → positively associated.

difference → removes variability

$\rho \approx 0.9016$

expla $\mathrm{Var}(T-N) = \mathrm{Var}(T) + \mathrm{Var}(N) - 2\,\mathrm{Cov}(T,N)$

import.

Q9

Diet	n	mean	SD
High-fat	8	28.4	2.1
control	10	25.9	3.0

ⓐ group 1 → high fat
group 2 → control

observed difference $= 28.4 - 25.9$
$= 2.5\ g$

$$a = \frac{\sigma_1^2}{n_1}$$

$$b = \frac{\sigma_2^2}{n_2}$$

$$a = \overset{0.55125}{(2.1)^2/8} \quad \text{and} \quad b = 3.0/10 = 0.9$$

$$SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$= \sqrt{a+b} = \sqrt{0.55125+0.9} = \sqrt{1.45} = 1.2042\ g$$

$$t = \frac{\mu_1 - \mu_2}{SE} = \frac{2.5}{1.2042} = 2.0761$$

ⓑ $$\nu = \frac{(a+b)^2}{\frac{a^2}{n_1-1} + \frac{b^2}{n_2-1}} = \frac{(a+b)^2}{\frac{a^2}{7} + \frac{b^2}{9}} = \frac{(0.55+0.9)^2}{\frac{(0.55)^2}{7} + \frac{(0.9)^2}{9}}$$

$$= \frac{(1.45)^2}{0.0432\ \frac{\cancel{0.3025}}{\cancel{7}} + \frac{\overset{0.09}{\cancel{0.81}}}{\cancel{9}}} = 2.1025$$

$$= \frac{2.1025}{0.0432 + 0.09}$$

$$= \frac{2.1025}{0.133} = 15.8083$$

$$\boxed{\nu = 15.80}$$

ⓒ Welch $t = 2.075$
$df = 15.8$
$P = 0.05468$

⑩ chi-square

observed table

	Broadly express	Not Broad	Total
CpG island	52	80-52=28	80
No CpG-island	28	70-28=42	70
total	80	70	150

$$\text{expected count} = \frac{\text{row total} \times \text{column}}{\text{grand total}}$$

$$\chi^2 = \frac{\Sigma(O-E)^2}{E} \qquad df = (\text{rows}-1)(\text{column}-1)$$

	Broadly express	Not Broad	Total
CpG island	80x80/150 = 42.67	80x70/150 = 37.33	80
No CpG-island	80x70/150 = 37.33	70x70/150 = 32.67	70

b) Test & conclusion

H_0 : promoter & expression → independant
H_1 : associated

$$\chi^2 : \frac{(52-42.67)^2}{42.67} + \frac{(28-37.33)^2}{37.33} + \frac{(28-37.33)^2}{37.33} + \frac{(42-32.67)^2}{32.67}$$

$= 9.375$

$df = 1$ (no correction)

$P = 0.00200$

$P < 0.05$ hence reject null hypothesis.
hence CpG promoter & expression is related

ⓒ fishers' exact test is used when

counts are too low for a reliable chi-square approx.

expected count below 5 is a warning

$OR = (52/28)/(28/42) = 2.786$

odds of broad expression is 2.786 times larger w/ CpG Island

Q11 ⓐ $z = \frac{\text{mean diff}}{\left(\frac{\sigma}{\sqrt{n}}\right)}$ under true effect

$$z \sim N\left(\frac{\delta\sqrt{n}}{\sigma}, 1\right)$$

$\sigma = 1.2 \quad \delta = 0.6 \quad n = 16$

$$m = \frac{\delta\sqrt{n}}{\sigma} = \frac{0.6\times4}{1.2} = 2$$

$$\text{Power} = P(Z > 1.96) + P(Z < -1.96) \rightarrow \text{2 sided}$$

$$= \phi(m - 1.96) + \phi(-m - 1.96)$$

$$= \phi(0.04) + \phi(-3.96) \rightarrow 0.00003747$$

approx

$$= 0.5159909$$

and design detect change of 0.6 only about 51.6%.

ⓑ $$\frac{\delta\sqrt{n}}{\sigma} - 1.96 \geq z_{80\%} = 0.8416$$

$$n \geq \left((1.96 + 0.8416)\frac{\sigma}{\delta}\right)^2$$

$$= \left[\frac{(2.8016)(1.2)}{0.6}\right] = 31.40$$

hence

required sample more than this i.e. 32

ⓒ with 20 changing gene we expect 20β to be changed

$$= 20(1 - 0.5160) = 20(0.484)$$

$$= 9.68$$

≈ 10 genes

⑫ Bonferroni → α/m

Holm $p_{(i)}$ $\frac{\alpha}{(m-i+1)}$ → stop @ 1st failure

BH largest k w/ $p_k \le \frac{kq}{m}$

ⓐ $P(\text{at least one } p < 0.05) = 1 - P(\text{all twelve} \ge 0.05)$

$= 1 - (0.95)^{12}$

$= 0.4596$

(b & c) Bonferroni threshold $\frac{0.05}{12} = \underline{0.004167}$

	p-val	Holm cutoff (b)		BH cutoff (c)	
1	0.0004	🟥 0.004167	reject	🟩 $\frac{1}{12} \times 0.05 = 0.004167$	
2	0.0021	🟥 0.05/11 = 0.0045		🟩 $\frac{2}{12} \times 0.05 = 0.00833$	
3	0.0039	🟥 $\frac{0.005}{10} = 0.005$		🟩 0.01250	
4	0.0052	🟥 $\frac{0.005}{9} = 0.0055$		🟩 0.01667	
5	0.011	0.006250	→ stop	🟩 0.02083	
6	0.018	🟩 0.007143	Accept	🟩 0.02500	
7	0.029	🟩 0.00833		🟩 0.02917	largest passing rank
8	0.041	🟩 0.0100		🟥 0.0333	
9	0.087	🟩 0.01250		🟥 0.03750	
10	0.21	🟩 0.01667		🟥 0.04167	
11	0.46	🟩 0.02500		🟥 0.04583	
12	0.73	🟩 0.050		🟥 0.05000	

green → Accepted

Red → rejected

ⓐ BH controls expected proportion of false discoveries among rejected hypothesis @ 5%

Bonferroni controls probability of any false rejection in family @ 5% without independence

Q13 Likelihood ratio

Posterior odds (A:B)
= Prior odds (A:B) × LR

odds = 0 $\quad \text{prob}(A) = \frac{0}{1+0}$ $\quad \frac{P(\text{data}/A)}{P(\text{data}/B)} =$

4 marker +ve

(A) $L(A) = C(5,4)\,(0.8)^4(0.2) = 0.4096$

$L(B) = C(5,4)(0.3)^4(0.7) = 0.02835$

$LR = \frac{0.4096}{0.02835} = 14.45$

observed count is 14.45x more likely to be under A

(B) Equal prior probabilities

$P(A) = P(B) = 0.5$

$P(\text{Data}/A) = 0.4096$

$P(D/B) = 0.02835$

$$P(A|D) = \frac{P(D/A) \times P(A)}{P(D)}$$

$P(D) = P(D/A)\,P(A) + P(D/B)\,P(B)$

$= (0.4096 \times 0.5) + 0.02835 \times 0.5$

$= 0.2048 + 0.01417$

$= 0.219$

$$P(A/D) = \frac{0.4096 \times 0.5}{0.219} = \frac{0.2048}{0.219} = 0.9351$$

ⓒ $^{5}C_{y} = 5$ is same as it does not depend on marker probabilities. So it cancels likelihood ratio and normalised posterior

ⓓ $P(A) = 0.1$ $P(B) = 0.9$

$A : B = 1 : 9$

Posterior odds $= \frac{1}{9}(14.448) = 1.6053$

$P(A|data) = \frac{1}{1.6053}(1 + 1.6053) = \frac{2.6053}{1.6053}$

$= 1.6229$

now 61.623 % posterior prob.

Q14 dispersion

$d = s^2/\text{mean}$

under poisson $X = \frac{(n-1)s^2}{\mu} = \chi^2_{n-1}$

negative binomial: $\text{var}(Y) = \mu + \phi\mu^2$

ⓐ mean $= \frac{112 + 87 + 140 + 95 + 166}{5}$

$= 600/5 = 120$

x	$x - \bar{x}$	$(x-\bar{x})^2$
112	−8	64
87	−33	1089
140	20	400
95	−25	625
166	46	2116
		4294

$\sigma^2 = \frac{4294}{4}$

$\sigma^2 = 1073.5$

$$D = \frac{\sigma^2}{\mu} = \frac{1073.5}{120}$$

$$\boxed{D = 8.946}$$

ⓑ H_0 : we assume poisson distributⁿ (x overdispersion)

H_1: counts exhibits overdispersion

$$X = \frac{(n-1)\sigma^2}{\mu}$$

$$= \frac{(5-1)(1073.5)}{120}$$

$$X = 35.783$$

under H_0 $X \sim \chi^2_{n-1} = \chi^2_4$

$$\chi^2_{0.95,4} = 9.488$$

$35.783 > 9.488$

hence we reject null hypothesis

ⓒ $\text{Var}(Y) = \mu + \phi\mu^2$

So

$$\hat{\phi} = \frac{\sigma^2 - \mu}{\mu^2}$$

$$\hat{\phi} = \frac{1073.5 - 120}{120^2}$$

$$= \frac{953.5}{14400}$$

$$\boxed{\hat{\phi} = 0.0662}$$

(a) Poission model assumes $\mu = \sigma^2$

but here the counts have much greater variance.

Q15

Genotype	Observed
AA	196
Aa	148
aa	56
Total	400

diploid → carries 2 allele

a) $p = \frac{2n_{AA} + n_{Aa}}{2N}$

$p = \frac{2(196) + 148}{2(400)}$

$p = 540/800$

$\boxed{p = 0.675}$

$\boxed{q = 1 - p = 0.325}$

ⓑ $p^2 + 2pq + q^2 = 1$

Genotype	Observed	Expected Obs.	Expected
AA	196	$400(0.675)^2$	182.25
Aa	148	$400(2)(0.675)(0.325)$	175.20
aa	56	$400(0.325)^2$	42.25
Total	400		

ⓒ H_0: population is hardy-weinberg eq^m

H_1: population deviates hardy-weinberg

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(196-182.25)^2}{182.25} + \frac{(148-175.5)^2}{175.5} + \frac{(56-42.25)^2}{42.25}$$

$$= 1.037 + 4.309 + 4.475$$

$\boxed{\chi^2 = 9.82}$ df $= 3-1-1=1$

$\chi^2_{0.95,1} = 3.841$

$9.82 > 3.841 \rightarrow$ reject H_0

ⓓ $F = 1 - \frac{\text{observed heterozygotes}}{\text{expected heterozygotes}}$

$= 1 - \frac{148}{175.5}$

$F \simeq 0.157$

15.7 % lower

Interpretation:

Q16

$X = 3, 1, 4, 2, 5, 1, 2, 2$

$P(X = x) = (1-p)^{x-1} \cdot p$

↳ prob. that progenitor

$n = 8 \quad \Sigma x_i = 20$

ⓐ $L(p) = \prod_{i=1}^{n} (1-p)^{x_i - 1} p$

$L(p) = p^n (1-p)^{\Sigma x_i - n}$

$\ell(p) = \ln L(p)$

$\ell(p) = n \ln p + (\Sigma x_i - n) \ln(1-p)$

for observation

$\ell(p) = 8 \ln p + 12 \ln(1-p)$

ⓑ $\ell'(p) = \frac{n}{p} - \frac{\Sigma x_i - n}{1-p}$

maximum-likelihood

$\frac{n}{p} = \frac{\Sigma x_i - n}{1-p}$

$$n(1-p) = p(\Sigma x_i - n)$$

$$n - np = p\Sigma x_i - np$$

$$n = p\,\Sigma x_i$$

Therefore $\hat{p} = \frac{n}{\Sigma x_i} \rightarrow \hat{p} = \frac{8}{20} = \frac{4}{10} = 0.4$

ⓒ $$l''(p) = \frac{-n}{p^2} - \frac{\Sigma x_i - n}{(1-p)^2}$$

$$\hat{p} = 0.4$$

$$l''(0.4) = \frac{-8}{0.4^2} - \frac{12}{0.6^2}$$

$$= -50 - 33.333$$

$$l''(0.4) = -83.33$$

Calculate → SE

$$SE(\hat{p}) \simeq \frac{1}{\sqrt{-l''(\hat{p})}}$$

$$SE(\hat{p}) = \frac{1}{\sqrt{83.33}}$$

$$SE(\hat{p}) \simeq 0.1095$$

ⓓ $$\hat{p} \pm \underset{\substack{\downarrow \\ 1.96}}{z_{0.975}}\, SE(\hat{p})$$

$$= 0.4 \pm 0.2147$$

95% CI = (0.185, 0.615)

ⓔ $E(x) = 1/p$ ∴ $\hat{p} = 0.4$

$E(x) \simeq 2.5$ divisions

17

Treated sample: $y_1 = 58$

Control sample: $y_2 = 34$

$Y_1 \sim \text{Poisson}(\lambda_1)$ $\quad H_0: \lambda_1 = \lambda_2$

$Y_2 \sim \text{Poisson}(\lambda_2)$ $\quad H_1: \lambda_1 \neq \lambda_2$

(a) $P(Y=y) = \frac{e^{-\lambda}\lambda^{y}}{y!}$

$$L(\lambda_1, \lambda_2) = \frac{e^{-\lambda_1}\lambda_1^{y_1}}{y_1!} + \frac{e^{-\lambda_2}\lambda_2^{y_2}}{y_2!}$$

$$\ell(\lambda_1, \lambda_2) = y_1 \ln \lambda_1 - \lambda_1 - \ln(y_1!) + y_2 \ln \lambda_2 - \lambda_2 - \ln(y_2!)$$

Unrestricted MLEs

$\hat{\lambda}_1 = y_1 = 58$

$\hat{\lambda}_2 = y_2 = 34$

$\hat{\lambda}_1 = 58 \qquad \hat{\lambda}_2 = 34$

Restricted MLE under H_0

$\lambda_1 = \lambda_2 = \lambda$

$\ell(\lambda) = (y_1 + y_2)\ln \lambda - 2\lambda - \ln(y_1!) - \ln(y_2!)$

$\frac{d\ell}{d\lambda} = \frac{y_1 + y_2}{\lambda} - 2$

$\frac{y_1 + y_2}{\lambda} - 2 = 0$

$\hat{\lambda} = \frac{y_1 + y_2}{2}$

$\hat{\lambda} = \frac{58 + 34}{2}$

$\boxed{\hat{\lambda} = 46}$

ⓑ $G = 2[\ell(\hat{\lambda}_1, \hat{\lambda}_2) - \ell(\hat{\lambda}, \hat{\lambda})]$

$$\ell(\hat{\lambda}_1, \hat{\lambda}_2) = y_1 \ln y_1 - y_1 - \ln(y_1!) + y_2 \ln y_2 - y_2 - \ln(y_2!)$$

$2\bar{y} = y_1 + y_2$

$$G = 2\left[y_1 \ln\left(\frac{y_1}{\bar{y}}\right) + y_2 \ln\left(\frac{y_2}{\bar{y}}\right)\right]$$

$$G = 2\sum_{j=1}^{2} y_i \ln\left(\frac{y_i}{\bar{y}}\right)$$

ⓒ Evaluate G and test @ the 5% level

$\bar{y} = 46$

$$G = 2\left[58 \ln\left(\frac{58}{46}\right) + 34 \ln\left(\frac{34}{46}\right)\right]$$

$$\boxed{G \simeq 6.334}$$

$G \sim \chi^2_1$

$df = 2 - 1 = 1$

$\chi^2_{0.95,1} = 3.841$

Compare

$6.334 > 3.841 \rightarrow$ reject H_0

$$\boxed{p \simeq 0.0118}$$

Conclusion — assumed poisson

$p < 0.05$ hence it is stastically significant

reject equality of Poisson mean @ 5% under assumed poisson model.

ⓐ The poison model assume

$Var(Y) = \mu$

$Var(Y) = \mu + \phi\mu^2$ (for -ve binomial overdisper)

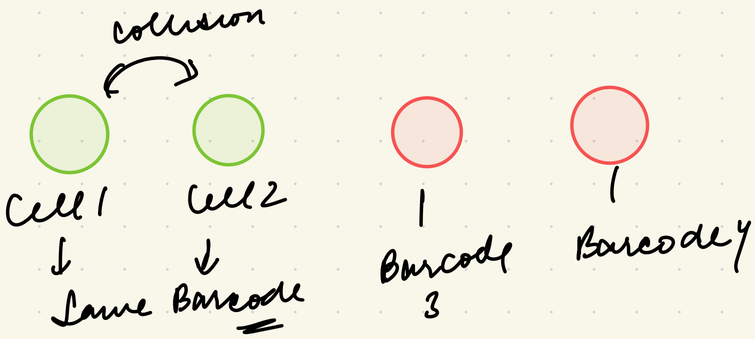
this is underestimated

so this can produce artifically small p-values.

So Poisson likehood ratio ynore biological over-dispersion and underegtmate variance, which leads to false-positive

Q18- coding question (attach later on github)

Q19


collision
cell 1
cell 2
same Barcode
Barcode 3
Barcode 4

ⓐ cell = 1000

we choose 2 to form collission

selection = ${}^{1000}C_2$

say cell 1 → receive barcode → 1 out of 100,000

for cell_2 to recieve same barcode is

$P(\text{same code}) = \frac{1}{100{,}000}$ P(collision for pair) = 10^{-5}

so we select in ${}^{1000}C_2$ ways and Prob. of them be same is 10^{-5}

$$E[K] = {}^{1000}C_2 \times 10^{-5} = 4.995 = \mu$$

ⓑ if colliding pair are poission

$$P(K=k) = \frac{e^{-\mu}\mu^k}{k!}$$

K: number of colliding pairs

k: no. of collision

μ = expected collision

$$P(X=0) = \frac{e^{-4.995}(4.995)^0}{0!}$$

$P(X=0) \simeq 0.00677$

P(no collision) = 0.677%

only 0.68%

ⓒ P(no. collision) ≥ 0.9

Barcode = n

$$\mu = 1000_{C_2}\left(\frac{1}{n}\right)$$

$$= \frac{499500}{n}$$

$$P(X=0) = \frac{e^{-\mu}\mu^0}{0!} = e^{-\mu} = e^{-\frac{499500}{n}}$$

$$e^{-\frac{499500}{n}} \geq 0.9$$

$$-\frac{499500}{n} \geq \ln(0.9)$$

$$-\frac{499500}{n} \geq -0.105$$

$$\frac{499500}{0.105} \leq n$$

$$4757142.86 \leq n$$

approximateley
4.76 million barcodes.